

Multi-Regime Multi-Fidelity Joint Model-Free Model-Based Diffusion for Soft Robot Co-design

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Abstract—Soft robot performance depends jointly on body morphology, material distribution, actuation, and control. This makes soft robot development a co-design problem, but the resulting search space is high-dimensional, contact-rich, and expensive to evaluate. Existing differentiable and generative co-design methods have shown that structured shape spaces and diffusion-based morphology priors can improve design search, but they typically assume a fixed simulation setting and do not explicitly reason about latent environment regimes or the cost of high-fidelity simulation. We propose a regime and fidelity-aware joint model-free model-based diffusion framework for soft robot co-design. A model-free morphology prior captures feasible topology and geometry, while model-based diffusion uses simulator feedback to refine controllers or designs through Monte Carlo score estimates. To improve robustness, we model contact and environment variation as latent regimes and derive a risk sensitive score estimator that emphasizes regimes where the current morphology-controller pair is likely to fail. To reduce simulation cost, we treat simulator fidelity as an adaptive decision variable inside the denoising process, selecting evaluations according to their estimated value for improving the final high-fidelity objective under a compute budget. We evaluate the method on soft robot locomotion and loco-manipulation tasks, demonstrating improved high-fidelity performance, robustness across environment regimes, and reduced high-fidelity simulation usage compared with fixed-fidelity and nominal co-design baselines.

I. INTRODUCTION

Soft robots derive their capability not only from control, but also from morphology. Their body geometry, material distribution, passive compliance, and actuator layout jointly determine how they move, contact the environment, and manipulate objects [1]–[4]. This makes soft robot design fundamentally different from conventional controller synthesis: a controller that is effective for one morphology may be ineffective for another, while a morphology that appears promising under one controller may fail under a different contact regime. Consequently, soft robot development is naturally a co-design problem, where morphology and control must be considered together rather than optimized in isolation.

Prior work has made substantial progress on computational soft robot design. Differentiable design pipelines [5] demonstrate that structured shape spaces and differentiable simulation can enable efficient shape–control optimization. Generative approaches further show that pretrained 3D diffusion models [6] can provide powerful morphology priors for exploring diverse soft robot designs. These works suggest that generative priors are valuable for searching over complex design spaces. However, existing approaches are less explicit about two issues that are central to practical soft robot co-

design: robustness to latent contact or environment regimes, and adaptive allocation of high-fidelity simulation budget.

Diffusion-based methods are attractive in this setting because they provide an iterative refinement mechanism over complex distributions [6]–[9]. A model-free diffusion prior can keep morphology search near feasible topology and geometry, while model-based diffusion can use simulator feedback to refine controllers or designs toward physically meaningful behavior. However, a naive diffusion-based co-design pipeline still faces two limitations: (1) if all environment variation is reduced to simple reward averaging, the resulting design may still fail in the regimes where it is most vulnerable; (2) if simulator fidelity follows a fixed coarse-to-fine schedule, the method assumes a useful scheduling pattern a priori, even though the best fidelity choice depends on the current denoising state, the reliability of low-fidelity estimates, and the remaining compute budget.

We propose a regime and fidelity-aware joint model-free model-based diffusion framework for soft robot co-design. Our method separates the roles of generative learning and physical optimization. A model-free morphology prior captures feasible topology and geometry, providing structured candidate bodies. Conditioned on these morphologies, model-based diffusion performs simulator-guided controller refinement using Monte Carlo score estimates. Rather than treating multi-fidelity simulation as a fixed coarse-to-fine schedule, we treat simulator fidelity as an adaptive decision variable inside the denoising process. At each step, the method decides whether a more expensive simulator evaluation is expected to improve the final high-fidelity objective enough to justify its cost. In parallel, environment uncertainty is modeled through a latent regime posterior, and robust guidance is obtained through a risk-sensitive score estimator that emphasizes failure-prone regimes.

The main contributions of this work are threefold:

- 1) We formulate soft robot co-design as a combination of learned morphology priors and simulator-guided model-based diffusion, allowing feasible geometry priors and physical task optimization to play complementary roles.
- 2) We treat simulator fidelity as an adaptive decision variable inside diffusion denoising rather than a fixed coarse-to-fine schedule, selecting fidelity based on its estimated value for improving the final high-fidelity objective under a compute budget.
- 3) We replace nominal contact-mode reward averaging with a risk-sensitive latent regime posterior, yielding a robust score estimator that focuses guidance on regimes in which

the current morphology–controller pair is most likely to fail.

II. RELATED WORK

A. Diffusion Methods for Soft Robot Co-Design

DiffuseBot [6] represents a more recent generative approach. Instead of optimizing only within a hand-designed parametric space, DiffuseBot leverages pretrained 3D shape diffusion models as structural priors and augments the generation process with physical simulation. Its robotizing procedure converts generated 3D shapes into soft robot morphologies, and differentiable simulation sensitivities are used to guide co-design. This shows that diffusion models can provide both diversity and useful structural bias for morphology search.

Our work builds on this insight but shifts the role of diffusion. Rather than using a generative model only to propose or refine morphology, we view soft robot co-design as a joint model-free model-based diffusion problem. The model-free component provides a topology and geometry prior over feasible soft bodies, while the model-based component performs online simulator-guided controller refinement. This separation allows us to focus on robustness and fidelity allocation, which are less explicit in existing generative co-design methods.

B. Uncertainty and Robustness in Robot Design and Control

Robots deployed in contact-rich environments must reason about uncertainty. In soft robotics, this uncertainty is not limited to sensor noise or parametric model error; it often appears as changes in contact regime, friction, object mass, terrain response, actuator behavior, or material deformation [10], [11].

In control, uncertainty-aware and risk-aware methods have been studied through robust MPC, probabilistic safety constraints, conformal prediction, and risk-sensitive optimization [12], [13]. Recent diffusion-based planning methods have also begun incorporating uncertainty or safety into denoising procedures, for example through uncertainty-aware dynamics models or safety-aware diffusion guidance [14], [15]. These works show that diffusion samplers can be modified to account for uncertainty, but they primarily address trajectory optimization rather than morphology–controller co-design.

Our work treats uncertainty as a latent environment regime rather than as a fixed nominal disturbance. The regime may represent contact condition, friction level, terrain type, object mass, or other environment-dependent effects. Instead of uniformly averaging rewards across regimes, we use a risk-sensitive latent-regime posterior to emphasize the regimes where the current design is most likely to fail. This gives the diffusion guidance a robustness objective that is more targeted than nominal reward optimization.

C. Multi-Fidelity Optimization for Robot Co-Design

Multi-fidelity optimization addresses the problem of making decisions when evaluations are available at different costs and accuracies [16], [17]. Prior work has used multi-fidelity evaluations, reduced-order models, transfer learning, surrogate models, and evolutionary search to reduce the cost of exploring

robot design spaces [18], [19]. These approaches demonstrate that low-cost evaluations can be useful for screening candidate designs before more expensive evaluations are applied. However, most existing multi-fidelity robot design methods are framed around evolutionary search, reinforcement learning, transfer learning, or surrogate-based optimization. They typically do not integrate fidelity selection into a diffusion denoising process, nor do they treat simulator fidelity as an adaptive decision variable coupled to score estimation.

In contrast, our method views each simulator fidelity as a biased and costly estimator of model-based guidance. The objective is not to follow a predetermined coarse-to-fine schedule, but to decide when a more expensive evaluation is valuable for improving the final high-fidelity outcome. This allows multi-fidelity simulation to become an adaptive component of model-based diffusion rather than a fixed experimental heuristic.

III. PRELIMINARIES

A. Problem Formulation

We consider soft robot co-design, where the goal is to find a morphology–controller pair that performs well under contact-rich dynamics and uncertain environment conditions. Let $x \in \mathcal{X}$ denote the robot morphology, including geometry, material distribution, and actuator layout, and let $\phi \in \Phi$ denote the controller parameters. We use $m \in \mathcal{M}$ to denote a latent environment regime, such as ground friction, object mass, object friction, terrain response, or contact condition. For a morphology–controller pair (x, ϕ) and regime m , the high-fidelity simulator returns a task reward $R_H(x, \phi; m)$.

The ideal robust co-design objective is

$$(x^*, \phi^*) = \arg \max_{x \in \mathcal{X}, \phi \in \Phi} \mathbb{E}_{m \sim p(m)} [R_H(x, \phi; m)]. \quad (1)$$

Directly solving Eq. (1) is difficult because morphology spaces are high-dimensional and structurally constrained, soft-body simulation is expensive, and the reward landscape changes across latent environment regimes.

To keep the notation general, we denote the diffusion variable by $z = (x, \phi)$. The following diffusion construction is written for a generic co-design variable z .

B. Joint Model-Free Model-Based Diffusion for Co-Design Sampling

We first introduce a joint model-free model-based sampling view for co-design. Unlike a purely model-free diffusion sampler, which only generates samples from a learned prior, or a purely model-based optimizer, which only refines variables through a simulator or optimization module, the joint view samples from both module spaces simultaneously and couples them through an interaction potential. In the soft robot co-design setting, we use this view to relate a model-free morphology module and a model-based controller refinement module.

Let z^{MF} denote the variable generated by the model-free module and z^{MB} denote the variable refined by the model-

based module. In our co-design setting, these variables are instantiated as

$$z^{\text{MF}} = x, \quad z^{\text{MB}} = \phi, \quad (2)$$

where x is the morphology and ϕ is the controller. More general instantiations may use a morphology latent for z^{MF} or include both morphology and controller components in each module.

The purpose of joint model-free/model-based sampling is to draw mutually compatible pairs

$$z = (z^{\text{MF}}, z^{\text{MB}}) \quad (3)$$

from a joint target distribution of the form

$$^1p(z^{\text{MF}}, z^{\text{MB}}) \propto p_\theta^{\text{MF}}(z^{\text{MF}}) p_0^{\text{MB}}(z^{\text{MB}}) V(z^{\text{MF}}, z^{\text{MB}}). \quad (4)$$

Here, p_θ^{MF} is the learned model-free diffusion prior, p_0^{MB} is the prior or proposal distribution of the model-based module, and V is an interaction potential. The interaction potential assigns high density to pairs for which the model-free output and the model-based output are compatible with each other and with the external objective. In co-design, this means that a generated morphology and a refined controller should jointly produce a physically successful soft robot behavior.

We now write the DDPM construction over the joint variable $z = (z^{\text{MF}}, z^{\text{MB}})$. Following standard DDPMs, let kz be the noisy joint variable at diffusion step k :

$$^kz = \sqrt{\bar{\alpha}_k} ^1z + \sqrt{1 - \bar{\alpha}_k} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \quad (5)$$

where $\bar{\alpha}_k := \prod_{j=1}^k \alpha_j$ and $^Kz \approx \mathcal{N}(0, \mathbf{I})$. The corresponding noisy marginal is

$$^kp(^kz) := \int q(^kz | ^1z) ^1p(^1z) d^1z. \quad (6)$$

The reverse-time denoising direction depends on the joint score

$$\mathbf{S}_k(^kz) := \nabla_{^kz} \log ^kp(^kz), \quad (7)$$

and the score-guided reverse update is

$$^{k-1}z = \frac{1}{\sqrt{\alpha_k}} \left(^kz + (1 - \bar{\alpha}_k) \hat{\mathbf{S}}_k \right), \quad (8)$$

where $\hat{\mathbf{S}}_k$ is an estimate of the joint score.

At reverse step k , the forward kernel induces a Gaussian proposal over clean joint candidates:

$$\tilde{z} = (\tilde{z}^{\text{MF}}, \tilde{z}^{\text{MB}}) \sim \Omega_k := \mathcal{N}\left(\frac{^kz}{\sqrt{\bar{\alpha}_k}}, \left(\frac{1}{\bar{\alpha}_k} - 1\right) \mathbf{I}\right). \quad (9)$$

Let

$$\Gamma_k = \{(\tilde{z}_m^{\text{MF}}, \tilde{z}_m^{\text{MB}})\}_{m=1}^M \quad (10)$$

be M samples from Ω_k . Each joint candidate is assigned an importance weight using the clean joint target:

$$\omega_m = ^1p(\tilde{z}_m^{\text{MF}}, \tilde{z}_m^{\text{MB}}) \propto p_\theta^{\text{MF}}(\tilde{z}_m^{\text{MF}}) p_0^{\text{MB}}(\tilde{z}_m^{\text{MB}}) V(\tilde{z}_m^{\text{MF}}, \tilde{z}_m^{\text{MB}}). \quad (11)$$

The self-normalized weights are

$$\bar{\omega}_m = \frac{\omega_m}{\sum_{j=1}^M \omega_j}. \quad (12)$$

The importance-weighted clean joint mean is then

$$^k\bar{z} = \sum_{m=1}^M \bar{\omega}_m \begin{bmatrix} \tilde{z}_m^{\text{MF}} \\ \tilde{z}_m^{\text{MB}} \end{bmatrix}. \quad (13)$$

This yields the Monte Carlo Score-Ascent estimator

$$\hat{\mathbf{S}}_k \approx -\frac{^kz}{1 - \bar{\alpha}_k} + \frac{\sqrt{\bar{\alpha}_k}}{1 - \bar{\alpha}_k} ^k\bar{z}. \quad (14)$$

Equations (8)–(14) define a joint sampling procedure: diffusion proposes model-free and model-based module outputs together, the interaction potential evaluates their compatibility, and the MCSA update moves the joint sample toward mutually compatible high-density pairs. This differs from a sequential pipeline in which a model-free sample is first generated and then independently corrected by a model-based module. In the joint view, the model-free and model-based variables are aligned through the same importance-weighted denoising step.

In our method, this generic joint sampling view is instantiated for soft robot co-design. The model-free variable represents morphology, the model-based variable represents the controller, and the interaction potential is constructed from simulator reward, latent-regime robustness, and multi-fidelity cost. The next subsection describes the corresponding co-design inference target.

C. Co-Design as Inference

We now define the target density $^1p(z)$ used by the model-based diffusion sampler. The central idea is to view co-design as inference: high-probability samples should correspond to co-design variables that are likely under the learned geometry prior and achieve high physical reward under simulator evaluation.

For the most general case, let $z = (x, \phi)$. We define the clean co-design target as

$$^1p(x, \phi) \propto p_\theta(x) p_0(\phi) \Psi(x, \phi), \quad (15)$$

where $p_\theta(x)$ is a model-free morphology prior, $p_0(\phi)$ is a controller prior, and $\Psi(x, \phi)$ is a model-based physical potential. The morphology prior encodes plausible topology and geometry, while the physical potential assigns high probability to designs and controllers that perform well in simulation.

When morphology is sampled from a fixed prior and model-based diffusion is used to refine only the controller, the conditional target becomes $^1p(\phi | x) \propto p_0(\phi) \Psi(x, \phi)$. This is the form used in our main implementation. It preserves the co-design structure because morphology candidates are generated from a learned morphology distribution, while their controllers are optimized through model-based denoising and the final morphology–controller pair is selected by high-fidelity performance.

A standard choice for the physical potential is the Boltzmann form [20], [21]

$$\Psi(x, \phi) = \exp\left(\frac{1}{\tau} R(x, \phi)\right), \quad (16)$$

where $\tau > 0$ is a temperature. Under this choice, samples with higher simulator reward receive larger importance weights

in Eq. (11), and therefore pull the denoising update toward physically better co-design variables.

In this work, the physical potential is generalized in two directions. First, the nominal reward $R(x, \phi)$ is replaced by a risk-sensitive latent-regime objective, so that importance weights emphasize regimes in which the current design is likely to fail. Second, the reward is evaluated using an adaptively selected simulator fidelity, so that the denoising process spends high-fidelity simulation only when its expected value justifies the cost. These two extensions modify the target density 1p used for importance weighting, while leaving the MBD sampling structure in Eq. (14) unchanged.

IV. METHODOLOGY

The preliminaries define MBD as an importance-weighted denoising procedure, where the sampler requires a clean target density $^1p(z)$ to reweight denoised candidates. We now instantiate this target for soft robot co-design. Our method has three components: an asymmetric model-free/model-based co-design pipeline, a latent-regime robust score estimator, and a budgeted multi-fidelity score guidance mechanism.

A. Latent-Regime Robust Score Estimation

The physical potential $\Psi(x, \phi)$ should not only reflect nominal task reward, because soft robot behavior is strongly affected by latent environment regimes. We therefore introduce a regime variable $m \in \mathcal{M}$, which may represent ground friction, object mass, object friction, terrain response, material response, or contact condition.

To emphasize failure-prone regimes, we define a risk-sensitive regime objective

$$\rho(x, \phi) = -\tau_r \log \sum_{m \in \mathcal{M}} p(m) \exp\left(-\frac{1}{\tau_r} R(x, \phi; m)\right), \quad (17)$$

where $\tau_r > 0$ controls the degree of risk sensitivity. As τ_r decreases, ρ approaches the worst-regime reward. The corresponding robust physical potential is

$$\Psi_{\text{risk}}(x, \phi) = \exp\left(\frac{1}{\tau} \rho(x, \phi)\right). \quad (18)$$

In the MBD sampler, this potential directly changes the importance weights assigned to denoised controller candidates:

$$\omega_m^k = ^1p(\tilde{\phi}_m | x) \propto p_0(\tilde{\phi}_m) \exp\left(\frac{1}{\tau} \rho(x, \tilde{\phi}_m)\right). \quad (19)$$

Therefore, the MCSA update is guided by samples that improve robust performance rather than only nominal or average performance.

B. Budgeted Multi-Fidelity Score Guidance

Evaluating the robust potential in Eq. (18) requires simulator rollouts. In soft-body co-design, high-fidelity simulation is expensive, while low-fidelity simulation is cheaper but biased. We therefore treat fidelity as an adaptive decision variable inside the denoising process rather than imposing a fixed coarse-to-fine schedule.

Algorithm 1 Regime- and Fidelity-Aware Model-Based Diffusion for Soft Robot Co-Design

Require: Morphology prior $p_\theta(x)$, controller prior $p_0(\phi)$, latent regimes $\mathcal{M}$, simulator fidelities $\{0, \dots, L\}$, fidelity costs $\{C_\ell\}$, diffusion steps K , proposal samples M , morphology candidates N_x , compute budget target $\bar{C}$

- 1: Initialize candidate set $\mathcal{C} \leftarrow \emptyset$.
 - 2: **for** $n = 1, \dots, N_x$ **do**
 - 3: Sample a morphology candidate $x_n \sim p_\theta(x)$.
 - 4: Robotize x_n into a simulatable soft-body design.
 - 5: **if** x_n fails geometric or simulation validity checks **then**
 - 6: continue.
 - 7: **end if**
 - 8: Initialize controller noise $^K\phi_n \sim \mathcal{N}(0, \mathbf{I})$.
 - 9: Initialize compute dual $\nu_K \geq 0$.
 - 10: **for** $k = K, K-1, \dots, 1$ **do**
 - 11: Construct the clean proposal from Eq. (9).
 - 12: Sample batch M candidates from Eq. (10).
 - 13: Estimate the value of each fidelity $\hat{\mathcal{V}}_k(\ell)$ using the current candidate batch and historical low/high-fidelity mismatch statistics.
 - 14: Select simulator fidelity Eq. (20).
 - 15: Evaluate candidate $\tilde{\phi}_m$ under selected fidelity ℓ_k^* and latent regimes $r \in \mathcal{M}$.
 - 16: Compute the objective from Eq. (21).
 - 17: Compute the importance weight Eq. (22).
 - 18: Compute the MCSA score estimate Eq. (14).
 - 19: Apply the score-guided reverse update
 - $$^{k-1}\phi_n = \frac{1}{\sqrt{\alpha_k}} \left(^k\phi_n + (1 - \bar{\alpha}_k) \hat{\mathbf{S}}_k \right).$$
 - 20: Update compute dual Eq. (23).
 - 21: **end for**
 - 22: **if** $(x_n, ^1\phi_n)$ passes high-fidelity certification **then**
 - 23: Add $(x_n, ^1\phi_n, R_H(x_n, ^1\phi_n))$ to $\mathcal{C}$.
 - 24: **end if**
 - 25: **end for**
 - 26: **Return** (x^*, ϕ^*)
-

Let $\ell \in \{0, \dots, L\}$ denote the simulator fidelity, with cost C_ℓ and reward $R_\ell(x, \phi; m)$. At diffusion step k , the fidelity is selected by a budgeted value criterion:

$$\ell_k^* = \arg \max_{\ell} \left[\hat{\mathcal{V}}_k(\ell) - \nu_k C_\ell \right], \quad (20)$$

where $\hat{\mathcal{V}}_k(\ell)$ estimates the value of using fidelity ℓ for improving the final high-fidelity objective, and ν_k is a compute-budget dual variable. This formulation does not assume that fidelity should monotonically increase with denoising time. A high-fidelity evaluation may be useful early if low-fidelity estimates mis-rank candidates, while low fidelity may remain sufficient late if candidate rankings are reliable.

Given the selected fidelity ℓ_k^* , the risk-sensitive reward

becomes

$$\rho_{\ell_k^*}(x, \phi) = -\tau_r \log \sum_{m \in \mathcal{M}} p(m) \exp \left(-\frac{1}{\tau_r} R_{\ell_k^*}(x, \phi; m) \right). \quad (21)$$

The resulting importance weight for a denoised controller candidate is

$$\omega_m^k \propto p_0(\tilde{\phi}_m) \exp \left(\frac{1}{\tau} \rho_{\ell_k^*}(x, \tilde{\phi}_m) - \nu_k C_{\ell_k^*} \right). \quad (22)$$

These weights are substituted into the same MCSA estimator.

The compute dual is updated according to the realized simulation cost:

$$\nu_{k+1} = [\nu_k + \eta_\nu (C_{\ell_k^*} - \bar{C})]_+, \quad (23)$$

where $\bar{C}$ is the desired average per-step cost. High-fidelity simulation is used in two roles: as a selective soft correction when its estimated value justifies the cost, and as a final hard certification signal for validating the returned morphology–controller pair.

C. Putting Everything Together

Algorithm 1 summarizes the proposed method. The algorithm follows the MBD sampling structure in Sec. III-B, but instantiates the clean target density using the co-design-specific physical potential defined in the previous subsections. A model-free morphology prior first proposes candidate bodies. For each morphology, model-based diffusion refines the controller by repeatedly sampling denoised controller candidates, evaluating them through simulator rollouts, constructing robust multi-fidelity importance weights, and applying the MCSA score update. The final morphology–controller pair is selected by high-fidelity validation.

V. EXPERIMENTAL ANALYSIS

We evaluate **M3BD** (Multi-Regime Multi-Fidelity Model-Based Diffusion) on a soft-voxel crawling co-design task to answer three questions:

- **Q1.** Does M3BD produce higher-quality morphology–controller pairs than sampling-based co-design baselines under a comparable evaluation budget?
- **Q2.** Does the latent-regime risk-sensitive score yield more robust designs across unseen contact regimes than mode-blind diffusion (MBD)?
- **Q3.** Does treating fidelity as an adaptive decision variable Pareto-dominate fixed single-fidelity schedules under the same compute budget?

Environment. The task is forward locomotion of a $4 \times 3 \times 4$ voxel-based soft body on a flat ground plane, simulated with a mass–spring soft-body engine. The morphology variable $x \in [0, 1]^{48}$ is a symmetry-tied stiffness map over voxels, and the controller $\phi \in \mathbb{R}^{8 \cdot T_a}$ parameterizes sine periodic actuator inputs on a fixed actuator basis. The reward $R(x, \phi; m)$ is the forward displacement of the body’s centroid after 200 control steps, evaluated under contact regime m parameterized by ground friction $\mu \in \{0.3, 0.4, 0.5, 0.6\}$.

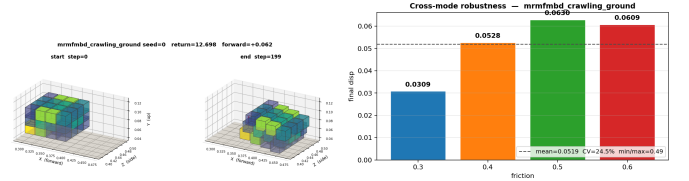


Fig. 1: **Best M3BD design (left)** at the start ($t=0$) and end ($t=199$) of an episode, showing forward displacement of 0.062. Voxel color denotes per-voxel stiffness. **Cross-mode robustness (right)** of the same design across four ground-friction regimes; the design generalizes to higher frictions and degrades gracefully on the low-friction regime.

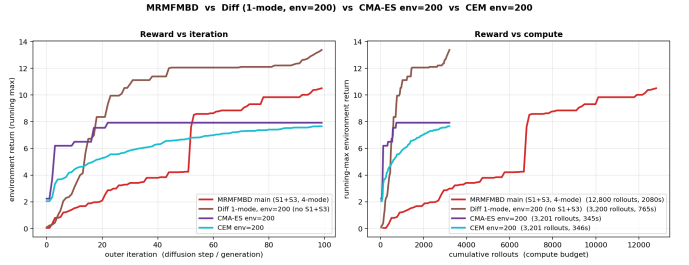


Fig. 2: **Q1: Reward curves for the four co-design methods.** Left: running-best return vs. outer iteration. Right: running-best return vs. cumulative simulator rollouts (compute budget). M3BD reaches a strictly higher final return than MBD, CMA-ES and CEM, and dominates throughout the second half of the budget.

Baselines. We compare M3BD against: (1) **CEM** [22], importance-weighted elite selection over the joint (x, ϕ) space; (2) **CMA-ES** [23], evolution-strategy search with a full covariance matrix; (3) **MBD** [8], model-based diffusion with single-mode reward averaging (no risk-sensitive regime potential, single fidelity); all run with the same morphology prior. We further include three single-fidelity ablations of M3BD that fix the simulator at $|m|=30, 100$, or 200 rollouts per evaluation, and a mode-blind variant that disables the regime potential while keeping the fidelity ladder.

Protocol. For each method we report (i) the in-distribution forward displacement $\bar{R}$ averaged over the four regimes, (ii) the cross-mode return at each held-out friction value, (iii) the worst-mode return $R_{\min}$, which directly measures regime-robustness, and (iv) the wall-clock cost. Diffusion methods use $K=100$ outer denoising steps and $S=53$ Monte Carlo controller samples per step; CEM and CMA-ES are matched to the same number of population generations. **Note that we don’t consider model-free prior in the experiment part, so all the methods are optimized starting from a simple uniform prior.**

A. Q1: Co-design Quality vs. Baselines

Table I reports per-regime cross-mode returns and the in-distribution mean. M3BD attains $\bar{R}=10.90$, improving over MBD (10.19, +7.0%), CMA-ES at the larger $|m|=200$ setting

Method	$\mu=0.3$	$\mu=0.4$	$\mu=0.5$	$\mu=0.6$	$\bar{R}$	$R_{\min}$	wall (s)
CEM ($ m =100$)	6.29	6.25	6.72	6.84	6.52	6.25	194
CMA-ES ($ m =100$)	5.49	5.95	6.31	6.21	5.99	5.49	175
CEM ($ m =200$)	4.21	7.50	7.45	6.81	6.49	4.21	347
CMA-ES ($ m =200$)	6.15	7.85	7.70	7.16	7.22	6.15	346
MBD (no regime)	5.31	11.35	12.62	11.47	10.19	5.31	523
M3BD (ours)	6.99	11.27	12.91	12.42	10.90	6.99	2081

TABLE I: Per-regime cross-mode return for the four methods (Q1), the worst-mode return $R_{\min}$ (Q2), and total wall-clock. Bold = best per column.

(7.92, +37.6%), and CEM at the same setting (7.66, +42.3%). The reward-vs-iteration and reward-vs-rollout curves in Fig. 2 make the gap concrete: CEM and CMA-ES plateau early because the elite selection collapses around a single mode of the joint morphology-controller space, whereas the diffusion-based methods continue to refine through later iterations as the noise schedule contracts. The wall-clock numbers in Table I also show that M3BD reaches a higher return at 2.08 ks than CEM and CMA-ES at 0.35 ks of single-fidelity $|m|=200$ rollouts: per-rollout, M3BD spends fewer rollouts on the final design and more on broad exploration, which is exactly what diffusion-style annealed inference provides.

B. Q2: Cross-Mode Robustness via the Regime Potential

The mode-blind MBD variant attains an in-distribution mean ($\bar{R}=10.19$) close to M3BD, but at the worst-case friction $\mu=0.3$ it collapses to $R_{\min}=5.31$, a 58% drop relative to its best-case mode. M3BD raises the worst-mode return to $R_{\min}=6.99$ (a +31.7% improvement over MBD on the metric that the risk-sensitive potential is designed to control), while only sacrificing 0.08 on $\mu=0.4$. The min/max ratio across regimes improves from 0.42 (MBD) to 0.54 (M3BD), so the regime-robust score does not just shift the mean but also flattens the regime profile. This is exactly the qualitative behavior predicted by Eq. (18): as τ_r shrinks, candidates that fail in any single regime are exponentially down-weighted, so the sampler is pushed away from brittle morphology-controller pairs.

Fig. 3 compares the reverse-diffusion trajectories of M3BD and MBD at six denoising steps. Both methods start from the same morphology prior at $k=99$ ($\text{env_return} \approx 0.04$) and end at $k=0$ with a clean design, but the trajectories differ in two ways. First, MBD commits earlier to a high-stiffness front block that produces a strong gait under the dominant high-friction regime ($k=19$, env_return 10.49), but this block is exactly what the low-friction regime cannot anchor against. Second, M3BD allocates more diffusion mass to morphologies with two compliant feet, which is a regime-hedging solution: it loses 0.5 at the in-distribution peak but gains 1.7 at the worst regime. The actuator activity panel below each morphology shows that the controller phase pattern becomes more periodic only in the last 20 steps, consistent with the picture of morphology converging first and controller fine-tuning happening as σ_k contracts.

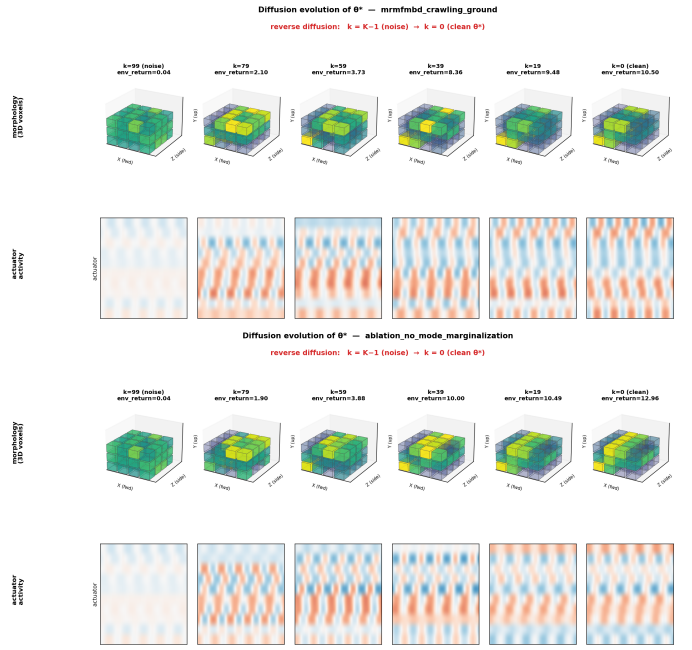


Fig. 3: **Reverse diffusion of $\theta = (x, \phi)$.** Top: M3BD with the regime-robust score. Bottom: MBD with single-mode averaging. Each row shows the morphology (top) and actuator activity (bottom) at $k \in \{99, 79, 59, 39, 19, 0\}$, annotated with the running env_return . M3BD ends at a slightly lower in-distribution return (10.50 vs. 12.96) but with a substantially higher worst-mode return (Tab. I).

C. Q3: Adaptive Fidelity vs. Fixed Schedules

Table II compares M3BD’s adaptive fidelity ladder ($|m| \in \{30, 100, 200\}$, selected by Eq. (20)) against fixed-fidelity ablations that use $|m|=30$, 100, or 200 rollouts at every diffusion step. Two patterns emerge.

(i) *Single-fidelity is bimodal.* Cheap evaluation ($|m|=30$) is fast (472 s) but biased: the score estimator mis-ranks candidates and the design saturates at $\bar{R}=7.74$. Expensive evaluation ($|m|=200$) recovers a strong design ($\bar{R}=12.29$) but at 3108 s, $1.5\times$ M3BD’s wall-clock. Intermediate $|m|=100$ is strictly worse than the two endpoints in return-per-second and is dominated by both.

(ii) *Adaptive ladder is on the Pareto frontier.* M3BD recovers 89% of the $|m|=200$ return at 67% of its wall-clock, and the worst-mode return $R_{\min}=6.99$ is close to the $|m|=200$ value of 8.33 at far lower cost. The right panel of Fig. 4 shows that the ladder dominates $|m|=30$ and $|m|=100$ over the entire compute budget, and only crosses below $|m|=200$ in the very last decile of the budget. Inspecting the dual-variable trace ν_k confirms the qualitative picture: M3BD spends low-fidelity rollouts when σ_k is large and the candidate ranking is noise-dominated, and switches to high-fidelity rollouts only after the morphology is mostly committed ($k < 40$). This is exactly the adaptive behavior that a fixed coarse-to-fine or fine-to-coarse schedule cannot express.

(iii) *Mode and fidelity are complementary.* Removing the

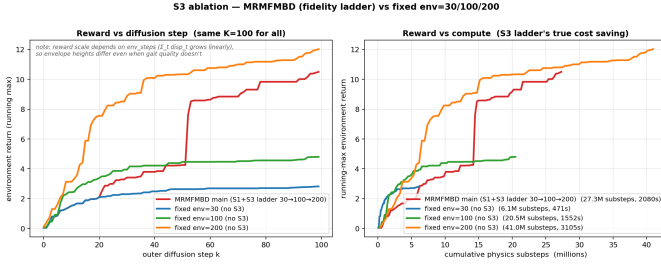


Fig. 4: **Q3: Fidelity ablation.** Reward vs. outer diffusion step (left) and reward vs. cumulative physics substeps (right) for M3BD’s adaptive ladder against fixed $|m|=30/100/200$ schedules. The ladder Pareto-dominates $|m|=30$ and $|m|=100$ across the full budget and matches $|m|=200$ at $\sim 2/3$ of its substep cost.

Variant	$\bar{R}$	$R_{\min}$	wall (s)	substeps
Fixed $ m =30$	7.74	7.66	472	6.1M
Fixed $ m =100$	8.72	7.57	1554	20.5M
Fixed $ m =200$	12.29	8.33	3108	41.0M
No regime, fixed $ m =200$	11.49	8.25	767	41.0M
M3BD adaptive (ours)	10.90	6.99	2081	27.3M

TABLE II: Fidelity ablation. “Adaptive” is the M3BD ladder over $|m| \in \{30, 100, 200\}$. “Fixed” rows replace the ladder by a constant $|m|$. The last row also disables the regime potential. $\bar{R}$ averages over $\mu \in \{0.3, 0.4, 0.5, 0.6\}$.

regime potential on top of the high-fidelity setting (*no mode, no fid.*, $|m|=200$) drops $\bar{R}$ from 12.29 to 11.49 and worsens $R_{\min}$ from 8.33 to 8.25. The two components contribute along different axes: the regime potential controls worst-mode behavior, while the fidelity ladder controls compute–quality tradeoff. Combining them gives the best return per second.

D. Discussion

The three experiments together show that the two design choices in M3BD are needed for different reasons. The regime-robust score (Q2) is what closes the worst-mode gap that mode-blind diffusion leaves open: the in-distribution mean is similar, but the brittle low-friction regime is where mode-blind methods quietly fail. The adaptive fidelity ladder (Q3) is what makes the method affordable: a sufficient single-fidelity baseline ($|m|=200$) exists, but only at $1.5\times$ the wall-clock and without any path to scaling the simulator further. We also note a limitation visible in Tab. II: the adaptive ladder’s $R_{\min}$ (6.99) is below the high-fidelity uniform baseline (8.33), meaning that the dual-variable schedule can occasionally allocate too few high-fidelity rollouts in regimes where the regime potential is most informative. A finer worst-regime-aware fidelity criterion is a natural next step.

VI. CONCLUSION

We proposed a regime and fidelity-aware joint model-free model-based diffusion framework for soft robot co-design. The method combines a learned morphology prior with simulator-weighted denoising, so that feasible robot bodies are proposed

from a model-free generative model while controllers are refined through model-based diffusion. The key idea is to make robustness and simulation cost part of the denoising process itself. Latent-regime weighting emphasizes failure-prone contact and environment conditions, while adaptive fidelity selection uses high-fidelity simulation only when it is expected to improve the final objective. This provides a unified way to search for morphology–controller pairs that are physically effective, more robust, and less expensive to evaluate.

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